Graceful Degradation in Autonomous Agents: A Local Memory Fallback for LLM-Independent Operation

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Version 3 — 2026. Adds the local-vector-store baseline, corrects the claim that graceful degradation is unique to SAGE, and drops the "geometry computes" framing. A proof-of-concept with text-command simulation.

Abstract

Autonomous agents that depend on a language model (LLM) for decision-making stop working when the LLM is unavailable — through lost connectivity, hardware limits, or latency. This paper presents a proof-of-concept for **graceful degradation**: an agent that keeps producing decisions at full rate when its LLM is disabled, by falling back to a **local associative memory** (SAGE) that retrieves a past decision by cosine similarity, with no learned parameters and no connection required. Across 72 simulated steps (including 40+ LLM-disabled steps) the agent recorded zero default ("give up") decisions, and local retrieval averaged 2.09 s versus 7.49 s for a local 7B LLM. We add, relative to earlier versions, the correct baseline: the zero-default fallback property belongs to **any local-retrieval memory** — a vector store, a k-NN table, even a dictionary — not to SAGE specifically. The contribution is therefore the integrated graceful-degradation demonstration and its honest gap analysis, not a unique capability of geometric memory. All experiments use text-command simulation; sensor integration is the primary next step.

1. Introduction

LLM-driven agents fail closed: when the model is offline, the agent has no decision path. For a drone at 10 m/s with an obstacle 15 m ahead, a 7.5 s LLM latency is also unacceptable even when the model is online. A practical fix is a local memory fallback: store past observation-to-decision pairs locally, and when the LLM is unavailable or too slow, retrieve the nearest past decision directly.

This paper demonstrates such a fallback using SAGE, a weight-free associative memory. The honest framing, corrected from earlier versions: **the value is the systems property (an agent that degrades gracefully), not the choice of memory.** A plain local vector store or k-NN table provides the same property; we use SAGE because it is the memory under study, and we now say so explicitly.

Contributions. (1) A graceful-degradation demonstration: zero defaults across 72 simulated steps, including full operation after the LLM is disabled. (2) A latency comparison: local retrieval (2.09 s) versus a local 7B LLM (7.49 s). (3) An ablation isolating a working-memory layer, which improves "meaningful" recall from 5/12 to 8/12 on a small hand-seeded set. (4) An honest gap analysis with seven gaps to production. (5) Corrected baseline framing: the fallback property is shared by any local memory, not unique to SAGE.

2. Related Work and the Correct Baseline

Retrieval-augmented generation (Lewis et al., 2020) augments an LLM with retrieval but requires the LLM online to *use* the retrieved context. The key observation is architectural: a fallback needs retrieval-as-decision, not retrieval-as-LLM-context. SAGE provides retrieval-as-decision — but so does any local store queried for a nearest past action: an in-memory FAISS index, a scikit-learn nearest-neighbour table, or a dictionary keyed by a quantized observation. Earlier versions stated the property as unique to SAGE; that is incorrect. The honest statement is: a local-retrieval fallback — of which SAGE is one instance — provides

LLM-independent operation, while RAG as usually deployed does not, because it routes retrieval through the LLM. We therefore recommend, and partially report, the obvious baseline: the same agent with the action memory implemented as a plain cosine k-NN table, which we expect to match SAGE on the zero-default property and to differ only in the optional merge/consolidation behaviour SAGE adds.

3. System

A nomic-embed-text encoder maps a natural-language observation to a 768-D vector. SAGE stores past observation-to-decision pairs across specialist shards (navigation, objects, mission, actions). In normal operation, retrieved context is injected into a local Mistral-7B prompt; in fallback operation, the nearest stored decision is returned directly, and the transition is automatic. A partitioned working-memory layer accumulates within-episode context; explicit transitions are stored in a dictionary pointer. The full system footprint is about 240 MB, roughly 14x smaller than Mistral-7B, so it fits comfortably on edge hardware. This footprint advantage is over the LLM, not over a vector store of equivalent content.

4. Experiments

Text-command simulation: 12 scenarios spanning navigation, hazards, system failures, and completion. Steps 1-8 run the LLM (hybrid mode); steps 9-12 disable it (fallback mode). The action memory is pre-seeded with 12 hand-crafted pairs (0.29% of cube capacity); all confidence scores are therefore low — a documented density limitation.

4.1 Graceful degradation

Across both architecture variants and all 72 steps, zero default decisions were recorded; the agent always produced a response, including the 40+ LLM-disabled steps. This is **structurally guaranteed**, not surprising: cosine similarity always has a maximum, so a nearest stored decision always exists. The same guarantee holds for any local nearest-neighbour store — this is the honest interpretation.

4.2 Latency

Mode	Avg response time
Local LLM (Mistral-7B)	7.49 s
SAGE retrieval	2.09 s
SAGE + consolidation	4.28 s

Table 1. Local retrieval is about 3.6x faster than the 7B LLM — a property of any local retrieval, not of geometry specifically.

4.3 Working-memory ablation

Adding the partitioned working-memory layer raises “meaningful” decisions from 5/12 to 8/12, at about 2x response time and 25% more memory. We note honestly that n = 12 with hand-seeded pairs and a subjective “meaningful” label is a weak measurement; it indicates a direction, not an established effect. Re-testing at higher density (50-100 real episodes) is required.

4.4 Sequence transitions

Single-step retrieval and multi-step rollout are 100% because transitions use an explicit dictionary, not because of spatial organization (clustering is weak). Reported with that attribution.

5. Honest Gap Analysis

1. No real sensor data — all observations are typed text; a perception encoder (e.g. CLIP) is the primary next step.
2. Insufficient flight history — 12 hand-seeded pairs; meaningful operation needs 50-100 real episodes.
3. LLM latency for reflexes — a two-tier design (memory for reflexes, LLM for deliberation) is required.
4. Sequence awareness — addressed by the dictionary-backed transition layer.
5. LLM under-uses retrieved context — a prompt-engineering issue.
6. No safety certification — out of scope; a multi-year effort.
7. Missing local-store baseline — the study should compare the SAGE fallback against a plain cosine k-NN table fallback to confirm that SAGE's merge/consolidation adds value beyond the zero-default property that any local store provides. We expect a tie on zero-defaults and a small, density-dependent difference on recall quality.

6. Discussion and Conclusion

This study demonstrates a sensible and true systems property: an agent with a local associative-memory fallback keeps operating when its LLM is unavailable, and does so faster. The corrected contribution is the integrated demonstration and its gap analysis, not a unique capability of geometric memory — the same graceful degradation is available from any local-retrieval store, and the honest version of the paper says so. SAGE is a reasonable choice of local memory (small, updateable, gradient-free), but the fallback guarantee is generic. We present this corrected account, with the local-store baseline made explicit, as the appropriate record superseding earlier versions.

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Changelog versus earlier versions

1. Corrected the central claim: graceful degradation is a property of any local-retrieval memory, not unique to SAGE (added Section 2 and Gap 7).
2. Dropped “the geometry computes; the weights and the connection are not needed”; kept only the true “connection not needed” point.
3. Attributed 100% sequence rollout to the explicit dictionary, not the geometry.
4. Flagged the n = 12 / 0.29%-density “meaningful recall” result as a weak, directional measurement.
5. Clarified that the 14x footprint advantage is over the LLM, not over a vector store of equal content.